

## The Collected Mathematical Papers of Arthur Cayley

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Papers 647 (conclusion), 648, 649, 650, 651, 652, 653, 654 (commencement)

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[647] ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION,  
 SYMMETRICAL DETERMINANT = 0 (CONCLUDED).

(Page 56.) The following table exhibits, for the surface in question, the incidence of the 8 nodes (numbered 1, ..., 8) with the 8 tropes (numbered I, ..., VIII):

|   | I | II | III | IV | V | VI | VII | VIII |
|---|---|----|-----|----|---|----|-----|------|
| 1 | • | .  | .   | .  | • | .  | •   | .    |
| 2 | . | •  | .   | •  | . | •  | .   | •    |
| 3 | . | .  | •   | .  | • | .  | •   | .    |
| 4 | . | •  | .   | •  | . | •  | .   | •    |
| 5 | • | .  | •   | .  | . | .  | •   | .    |
| 6 | . | •  | .   | •  | . | •  | .   | .    |
| 7 | . | .  | •   | .  | • | .  | .   | •    |
| 8 | . | •  | .   | .  | . | .  | •   | •    |

viz. reading horizontally or vertically, the dots show the tropes through each node, or the nodes in each trope.

The plane through any ray and the nodal line is a pinch-plane of the surface, its point of contact being the intersection of the ray with the nodal line; and the intersection of each axis with the nodal line is a pinch-point of the surface, the tangent plane being the plane through the axis and the nodal line; the surface has thus 4 pinch-planes and 4 pinch-points.

## 648.

## ALGEBRAICAL THEOREM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), p. 53.]

(Page 57.) I wish to put on record the following theorem, given by me as a Senate-House Problem, January, 1851.

If  $\{\alpha + \beta + \gamma + \dots\}^p$  denote the expansion of  $(\alpha + \beta + \gamma + \dots)^p$ , retaining those terms  $N\alpha^a\beta^b\gamma^c\dots$  only in which

$$b + c + d + \dots > p - 1, \quad c + d + \dots > p - 2, \quad \&c., \&c.,$$

then

$$\begin{aligned} x^n = (x + \alpha)^n - n\{\alpha\}^1(x + \alpha + \beta)^{n-1} + \frac{1}{2}n(n-1)\{\alpha + \beta\}^2(x + \alpha + \beta + \gamma)^{n-2} \\ - \frac{1}{6}n(n-1)(n-2)\{\alpha + \beta + \gamma\}^3(x + \alpha + \beta + \gamma + \delta)^{n-3} + \&c. \end{aligned}$$

The theorem, in a somewhat different and imperfectly stated form, is given, Burg, *Crelle*, t. i. (1826), p. 368, as a generalisation of Abel's theorem,

$$\begin{aligned} (x + \alpha)^n = x^n + n\alpha(x + \beta)^{n-1} + \frac{1}{2}n(n-1)\alpha(\alpha - 2\beta)(x + 2\beta)^{n-2} \\ + \frac{1}{6}n(n-1)(n-2)\alpha(\alpha - 3\beta)^2(x + 3\beta)^{n-3} + \&c. \end{aligned}$$

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8

## 649.

**ADDITION TO MR GLAISHER'S NOTE ON SYLVESTER'S PAPER,  
"DEVELOPMENT OF AN IDEA OF EISENSTEIN."**

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 83, 84.]

(Page 58.) The formula (11) [in the Note], under a slightly different form, is demonstrated by me in an addition [263] to Sir J. F. W. Herschel's paper "On the formulae investigated by Dr Brinkley, &c.," *Phil. Trans.*, t. CL., 1860, pp. 321–323. The demonstration is in effect as follows: let  $u$  denote a series of the form  $1 + bx + cx^2 + dx^3 + \dots$ , and let  $u^i$  (where  $i$  is positive or negative, integer or fractional) denote the development of the  $i$ -th power of  $u$ , continued up to the term which involves  $x^n$ , the terms involving higher powers of  $x$  being rejected;  $u^0, u^1, u^2$ , and generally  $u^i$  will denote in like manner the developments of these powers up to the terms involving  $x^n$ , or, what is the same thing, they will be the values of  $u^i$  corresponding to  $i = 0, 1, 2, \dots, s$ . By the formula

$$u^i = 1 + \frac{i}{1}(u-1) + \frac{i \cdot i - 1}{1 \cdot 2}(u-1)^2 + \dots \text{ as far as the term involving } (u-1)^n, \text{ } u^i \text{ is a rational}$$

and integral function of  $i$  of the degree  $n$ , and can therefore be expressed in terms of the values  $u^0, u^1, u^2, \dots, u^n$  which correspond to  $i = 0, 1, 2, \dots, n$ . Let  $s$  have any one of the last-mentioned values, then the expression

$$\frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)},$$

which as regards  $i$  is a rational and integral function of the degree  $n$  (the factor  $i - s$  which occurs in the numerator and denominator being of course omitted), vanishes for each of the values  $i = 0, 1, 2, \dots, n$ , except only for the value  $i = s$ , in which case it becomes equal to unity. The required formula is thus seen to be

$$u^i = \sum \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)} u^s \right\},$$

(Page 59.) where the summation extends to the several values  $s = 0, 1, 2, \dots, n$ ; or, what is the same thing, it is

$$u^i = \sum \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{(-)^{n-s}}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots (n - s)} u^s \right\},$$

or, changing the sign of  $i$ , it is

$$u^{-i} = \sum \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^n}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} u^s \right\},$$

where, as before,  $s$  has the values  $0, 1, 2, \dots, n$  successively. Or, what is the same thing, we have

$$C_{-i,n} = \sum \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^n}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} C_{s,n} \right\},$$

where the term corresponding to  $s = 0$ , as containing the factor  $C_{0,n}$  vanishes except in the case  $n = 0$  (for which it is  $= 1$ ); and omitting this evanescent term, this is in fact the formula (11).

650.

ON A QUARTIC SURFACE WITH TWELVE NODES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 103–106.]

(Page 60.) Write for shortness

$$\begin{array}{lll} a = \beta - \gamma, & f = \alpha - \delta, & af = p, \\ b = \gamma - \alpha, & g = \beta - \delta, & bg = q, \\ c = \alpha - \beta, & h = \gamma - \delta, & ch = r; \end{array}$$

then,  $\theta$  being a variable parameter, the surface in question is the envelope of the quadric surface

$$(\alpha + \theta)^2 agh X^2 + (\beta + \theta)^2 bhf Y^2 + (\gamma + \theta)^2 cfg Z^2 + (\delta + \theta)^2 abc W^2 = 0;$$

viz. this is

$$\Sigma \theta^2 agh X^2 \cdot \Sigma agh X^2 - \Sigma \alpha agh X^2 = 0.$$

There are no terms in  $X^4$ , &c.; the coefficient of  $Y^2 Z^2$  is

$$\gamma^2 cfg \cdot bfh + \beta^2 bfh \cdot cfg - 2\beta\gamma bfh \cdot cfg,$$

which is

$$= bc fgh (\beta - \gamma)^2, \quad = a^2 bc f^2 gh, \quad = abc fgh \cdot p.$$

Hence the whole equation divides by  $abc fgh$ , and throwing out this factor, the result is

$$p(Y^2 Z^2 + X^2 W^2) + q(Z^2 X^2 + Y^2 W^2) + r(X^2 Y^2 + Z^2 W^2) = 0,$$

or, observing that  $p + q + r = 0$ , this may also be written

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2 = 0,$$

(Page 61.) and also

$$p(YZ - XW)^2 + q(ZX - YW)^2 + r(XY - ZW)^2 = 0.$$

The more general equation

$$(p, q, r, l, m, n \mid YZ + XW, ZX + YW, XY + ZW)^2 = 0$$

represents a quartic surface (octadic) having the 8 nodes

$$\begin{array}{ll} (1, 0, 0, 0), & (\bar{1}, 1, 1, 1), \\ (0, 1, 0, 0), & (1, \bar{1}, 1, 1), \\ (0, 0, 1, 0), & (1, 1, \bar{1}, 1), \\ (0, 0, 0, 1), & (1, 1, 1, \bar{1}). \end{array}$$

We have

| $d_x U =$                      | $d_y U =$                      |
|--------------------------------|--------------------------------|
| $p. \quad XW^2 + YZW$          | $p. \quad YZ^2 + XZW$          |
| $q. \quad YW^2 + YZW$          | $q. \quad YW^2 + XZW$          |
| $r. \quad ZW^2 + YZW$          | $r. \quad YX^2 + XZW$          |
| $l. \quad 2XYZ + W(Y^2 + Z^2)$ | $l. \quad 2XYW + Z(W^2 + X^2)$ |
| $m. \quad 2XYW + Z(W^2 + Y^2)$ | $m. \quad 2YZX + W(Z^2 + X^2)$ |
| $n. \quad 2XZW + Y(W^2 + Z^2)$ | $n. \quad 2YZW + X(W^2 + Z^2)$ |

| $d_z U =$                      | $d_w U =$                       |
|--------------------------------|---------------------------------|
| $p. \quad Y^2 Z + XYW$         | $p. \quad X^2 W + XYZ$          |
| $q. \quad X^2 Z + XYW$         | $q. \quad Y^2 W + XYZ$          |
| $r. \quad W^2 Z + XYW$         | $r. \quad Z^2 W + XYZ$          |
| $l. \quad 2ZXW + Y(W^2 + X^2)$ | $l. \quad 2WYZ + X(Y^2 + Z^2)$  |
| $m. \quad 2YZW + X(W^2 + Y^2)$ | $m. \quad 2WZX + Y(Z^2 + X^2)$  |
| $n. \quad 2ZXY + W(X^2 + Y^2)$ | $n. \quad 2WXY + Z(X^2 + Y^2).$ |

Hence there will be a node

$$\begin{aligned}
 1, 1, 1, 1, \quad & \dots p + q + r + 2l + 2m + 2n = 0, \\
 \bar{1}, 1, 1, 1, \quad & \dots p + q + r - 2l + 2m - 2n = 0, \\
 1, \bar{1}, 1, 1, \quad & \dots p + q + r - 2l + 2m - 2n = 0, \\
 \bar{1}, \bar{1}, 1, 1, \quad & \dots p + q + r - 2l - 2m + 2n = 0, \\
 1, 1, \bar{1}, 1, \quad & \dots p + q + r + 2l + 2m + 2n = 0;
 \end{aligned}$$

or say there will be

$$\begin{aligned}
 1 \text{ of these nodes if } & p + q + r + 2l + 2m + 2n = 0, \\
 2 \dots & p + q + r + 2l = 0, \quad m + n = 0, \\
 3 \dots & p + q + r = 2l = -2m = -2n, \\
 4 \dots & p + q + r = 0, \quad l = 0, \quad m = 0, \quad n = 0;
 \end{aligned}$$

(Page 62.) viz. the surface having the 12 nodes is the original surface

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2,$$

where

$$p + q + r = 0.$$

The Jacobian of the quadrics

$$YZ + XW = 0, \quad ZX + YW = 0, \quad XY + ZW = 0,$$

is

$$\begin{vmatrix}
 W, & Z, & Y, & X \\
 Z, & W, & X, & Y \\
 Y, & X, & W, & Z \\
 X, & Y, & Z, & W
 \end{vmatrix} = 0;$$

viz. the equations are

$$\begin{aligned}
 X^3 - X(Y^2 + Z^2 + W^2) + 2YZW &= 0, \\
 Y^3 - Y(Z^2 + X^2 + W^2) + 2ZXW &= 0, \\
 Z^3 - Z(X^2 + Y^2 + W^2) + 2XYW &= 0, \\
 W^3 - W(X^2 + Y^2 + Z^2) + 2XYZ &= 0,
 \end{aligned}$$

each of which is satisfied in virtue of any one of the pairs of equations

$$\begin{aligned}
 (Y - Z = 0, \quad X - W = 0) \quad & (Y + Z = 0, \quad X + W = 0), \\
 (Z - X = 0, \quad Y - W = 0) \quad & (Z + X = 0, \quad Y + W = 0), \\
 (X - Y = 0, \quad Z - W = 0) \quad & (X + Y = 0, \quad Z + W = 0).
 \end{aligned}$$

So that the Jacobian curve is, in fact, the six lines represented by these equations.

Any two of the three tetrads form an octad, the 8 points of intersection of three quadric surfaces: a figure representing the relation of the 12 points to each other may be constructed without difficulty.

Each tetrad is a sibi-conjugate tetrad *quoad* the quadric  $X^2 + Y^2 + Z^2 + W^2 = 0$ . The three tetrads are not on the same quadric surface.

## 651.

## ON A SPECIAL SURFACE OF MINIMUM AREA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 190–196.]

(Page 63.) A very remarkable form of the surface of minimum area was obtained by Prof. Schwarz in his memoir “Bestimmung einer speciellen Minimal-fläche,” Berlin, 1871, [*Ges. Werke*, t. 1., pp. 6–125], crowned by the Academy of Sciences at Berlin. The equation of the surface is

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where  $\lambda, \mu, \nu$  are functions of  $x, y, z$  respectively, viz.

$$x = - \int_0^\lambda \frac{d\theta}{\sqrt{(\frac{1}{4}\theta^4 + \frac{1}{2}\theta^2 + \frac{1}{4})}},$$

and  $y, z$  are the same functions of  $\mu, \nu$  respectively. A direct verification of the theorem that this is a surface of minimum area, satisfying, that is, the differential equation

$$r(1 + q^2) - 2pqs + t(1 + p^2) = 0,$$

is given in the memoir; but the investigation may be conducted in quite a different manner, so as to be at once symmetrical and somewhat more general, viz. we may enquire whether there exists a surface of minimum area

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where the determining equations are

$$\lambda'^2 = a\lambda^4 + b\lambda^2 + c,$$

$$\mu'^2 = a\mu^4 + b\mu^2 + c,$$

$$\nu'^2 = a\nu^4 + b\nu^2 + c,$$

(Page 64.)  $\left(\lambda' = \frac{d\lambda}{dx}, \text{ \&c.}\right)$ . I find that the coefficients  $a, b, c$  must satisfy four homogeneous quadric equations, which, in fact, admit of simultaneous solution, and that in three distinct ways; viz. assuming  $a = 1$ , the solutions are

$$\begin{aligned} a = 1, \quad b = \frac{1}{2}, \quad c = 1, \\ a = 1, \quad b = -2, \quad c = 1, \\ a = 1, \quad b = -\frac{1}{2}, \quad c = -\frac{1}{2}; \end{aligned}$$

that is,

$$\lambda'^2 = \lambda^4 + \frac{1}{2}\lambda^2 + 1 \left\{ = \frac{1}{4}(4\lambda^4 + 2\lambda^2 + 4) \right\},$$

which gives Schwarz's surface;

$$\lambda'^2 = \lambda^4 - 2\lambda^2 + 1 \text{ or } \lambda' = \pm(\lambda^2 - 1),$$

which, it is easy to see, gives only  $x + y + z = \text{const.}$ ; and

$$\lambda'^2 = \lambda^4 - \frac{1}{2}\lambda^2 - \frac{1}{2}, \quad = (\lambda^2 - 1)(\lambda^2 + \frac{1}{2}),$$

which is a surface similar in its nature to Schwarz's surface.

The investigation is as follows: the condition to be satisfied by a surface of minimum area  $U = 0$  is

$$(\mathfrak{a} + \mathfrak{b} + \mathfrak{c})(X^2 + Y^2 + Z^2) - (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid X, Y, Z)^2 = 0,$$

where  $(X, Y, Z)$  are the first derived coefficients and  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h})$  the second derived coefficients of  $U$  in regard to the coordinates. Considering  $U$  as a function of  $\lambda, \mu, \nu$ , which are functions of  $x, y, z$  respectively, and writing  $(L, M, N)$  and  $(a, b, c, f, g, h)$  for the first and second derived functions of  $U$  in regard to  $\lambda, \mu, \nu$ , also  $\lambda', \lambda''$  for the first and second derived functions of  $\lambda$  in regard to  $x$ , and so for  $\mu, \mu''$  and  $\nu, \nu''$ : we have

$$(X, Y, Z) = (L\lambda', M\mu', N\nu'),$$

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}) = (a\lambda'^2 + L\lambda'', b\mu'^2 + M\mu'', c\nu'^2 + N\nu'', f\mu'\nu', g\nu'\lambda', h\lambda'\mu'),$$

and for the particular surface  $U = 1 + \mu\nu + \nu\lambda + \lambda\mu = 0$ , the values are

$$(L, M, N, a, b, c, f, g, h) = (\mu + \nu, \nu + \lambda, \lambda + \mu, 0, 0, 0, 1, 1, 1).$$

Hence the condition is found to be

$$\begin{aligned} & 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) \\ & + 2\nu'^2\lambda'^2(\mu + \nu)(\mu + \lambda) \\ & + 2\lambda'^2\mu'^2(\nu + \lambda)(\nu + \mu) \\ & - \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} \\ & - \mu''(\nu + \lambda)\{(\mu + \lambda)^2\nu'^2 + (\mu + \nu)^2\lambda'^2\} \\ & - \nu''(\lambda + \mu)\{(\nu + \mu)^2\lambda'^2 + (\nu + \lambda)^2\mu'^2\} = 0, \end{aligned}$$

(Page 65.) or say this is

$$\Sigma 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) - \Sigma \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} = 0.$$

We have to write in this equation  $\lambda'^2 = a\lambda^4 + b\lambda^2 + c$ , and therefore  $\lambda'' = 2a\lambda^3 + b\lambda$ , &c.; the left-hand side, call it  $\Omega$ , is a symmetrical function of  $\lambda, \mu, \nu$ , and is consequently expressible as a rational function of

$$\begin{aligned} p &= \lambda + \mu + \nu, \\ q &= \mu\nu + \nu\lambda + \lambda\mu, \\ r &= \lambda\mu\nu. \end{aligned}$$

We ought to have  $\Omega = 0$ , not identically, but in virtue of the equation  $1 + q = 0$ , that is,  $\Omega$  should divide by  $1 + q$ ; or, what is the same thing,  $\Omega$  should vanish on writing therein  $q = -1$ .

To effect the reduction as easily as possible, observe that we have  $(\lambda + \mu)(\lambda + \nu) = \lambda^2 + q$ ; and therefore

$$\Sigma \mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) = \Sigma \lambda^2\mu'^2\nu'^2 + q\Sigma \mu'^2\nu'^2.$$

Similarly, in the second term,

$$(\mu + \nu)(\lambda + \nu)^2 = (\nu + \lambda)(\nu^2 + q) \quad \text{and} \quad (\mu + \nu)(\lambda + \mu)^2 = (\mu + \lambda)(\mu^2 + q).$$

The complete value of  $\Omega$  thus is

$$\Omega = 2(Aq + B) - [(C + D)q + E + F],$$



where

$$\begin{aligned} A &= \Sigma \lambda^2 \mu'^2 \nu'^2, & B &= \Sigma \mu'^2 \nu'^2, \\ C &= \Sigma \lambda \lambda'' (\nu \mu'^2 + \mu \nu'^2), & D &= \Sigma \lambda'' (\mu^2 \nu'^2 + \nu^2 \mu'^2), \\ E &= \Sigma \lambda \lambda'' (\mu'^2 + \nu'^2), & F &= \Sigma \lambda'' (\nu \mu'^2 + \mu \nu'^2). \end{aligned}$$

We find without difficulty

$$\begin{aligned} A &= a^2 (q^4 - 4q^2 pr + 4qr^2 + 2p^2 r^2) \\ &\quad + ab (-2q^3 + qp^2 + 4qpr - 3r^2 - 2p^2 r) \\ &\quad + ac (4q^2 - 8qp^2 + 8pr + 2p^4) \\ &\quad + b^2 (q^2 - 2pr) \\ &\quad + bc (-4q + 2p^2) \\ &\quad + c^2 (3), \\ B &= a^2 (q^2 r^2 + 2pr^3) \\ &\quad + ab (-4qr^2 + 2p^2 r^2) \\ &\quad + ac (-2q^3 + q^2 p^2 + 4qpr - 3r^2 - 2p^2 r) \\ &\quad + b^2 (3r^2) \\ &\quad + bc (2q^2 - 4pr) \\ &\quad + c^2 (-2q + 2p^2), \end{aligned}$$

(Page 66.)

$$\begin{aligned} C &= a^2 ( ) \\ &\quad + ab (-6q^3 + 3q^2 p^2 + 12qpr - 9r^2 - 6p^2 r) \\ &\quad + ac (8q^2 - 16qp^2 + 16pr + 4p^4) \\ &\quad + b^2 (2q^2 - 4pr) \\ &\quad + bc (4q + 2p^2), \\ D &= a^2 (4q^2 pr - 2qr^2 - 4p^2 r^2) \\ &\quad + ab (-4qpr + 2p^2 r) \\ &\quad + ac (-4q^2 + 2qp^2 - 2pr) \\ &\quad + b^2 (2pr) \\ &\quad + bc (2q), \\ E &= a^2 (4q^2 r^2 - 4pr^3) \\ &\quad + ab (-12qr^2 + 6p^2 r^2) \\ &\quad + ac (-4q^3 + 2q^2 p^2 + 8qpr - 6r^2 - 4p^2 r) \\ &\quad + b^2 ( ) \\ &\quad + bc (2q^2 - 4pr), \\ F &= a^2 ( ) \\ &\quad + ab ( ) \\ &\quad + ac (4q^3 - 12qpr + 12p^2 r) \\ &\quad + b^2 (qpr - 3r^2) \\ &\quad + bc (-2q^2 + qp^2 - pr), \end{aligned}$$

where in each line the terms are arranged according to their order in  $p, r$ .

Substituting, we find

$$\begin{aligned}\Omega = & a^2(-2q^3 + 6q^2pr - 8qr^2) \\ & + ab(2q^2 - q^3pr - 4qr^2) \\ & + ac(-2q^2p^2 + 14qpr - 12r^2) \\ & + b^2(-3qpr + 3r^2) \\ & + bc(-2q^2 + qp^2 - 3pr) \\ & + c^2(2q + 2p^2); \end{aligned}$$

viz. writing  $q = -1$ , this is

$$\begin{aligned}\Omega = & a^2(2 - 6pr - 8r^2) \\ & + ab(2 + p^2 - pr - 4r^2) \\ & + ac(-2p^2 - 14pr - 12r^2) \\ & + b^2(3pr + 3r^2) \\ & + bc(-2 - p^2 - 3pr) \\ & + c^2(-2 - 2p^2); \end{aligned}$$

(Page 67.) or, what is the same thing, it is

$$\begin{aligned}\Omega = & (2a^2 + 2ab - 2bc - 2c^2) \\ & + p^2(ab - 2ac - bc + 2c^2) \\ & + pr(-6a^2 - ab - 14ac + 3b^2 - 3bc) \\ & + r^2(-8a^2 - 4ab - 12ac + 3b^2); \end{aligned}$$

so that, writing for convenience  $a = 1$ , the equations to be satisfied are

$$\begin{aligned}2 - 2c^2 + 2(1 - c)b &= 0, \\ -2c + 2c^2 + (1 - c)b &= 0, \\ -6 - 14c + 3b^2 - (1 + 3c)b &= 0, \\ -8 - 12c + 3b^2 - 4b &= 0. \end{aligned}$$

The first and second are  $(1 - c)(2 + 2c + 2b) = 0$  and  $(1 - c)(-2c + b) = 0$ ; viz. they give  $c = 1$ , or else  $b = -2$ ,  $c = -\frac{1}{2}$ . In the former case, the third and fourth equations each become  $3b^2 - 4b - 20 = 0$ , that is  $(3b - 10)(b - 2) = 0$ ; in the latter case, they are satisfied identically; hence we have for  $a, b, c$  the three systems of values mentioned at the beginning.

This completes the investigation; but it is interesting to find the values assumed by the other factor of  $\Omega$  on substituting therein for  $a, b, c$  the foregoing several systems of values. We have

in general

$$\begin{aligned}
\Omega &= -2a^2q^3 + 2abq^2 - 2bcq + 2c^2q \\
&\quad + p^2(-abq^2 - 2acq^2 + bcq + 2c^2) \\
&\quad + pr(6a^2q^2 - abq + 14acq - 3b^2q - 3bc) \\
&\quad + r^2(-8a^2q + 4abq - 12ac + 3b^2) \\
&= -2a^2(q^3 - 1) + 2ab(q^2 - 1) - 2bc(q - 1) + 2c^2(q + 1) \\
&\quad + p^2\{-ab(q^2 + 1) - 2ac(q^2 - 1) + bc(q + 1)\} \\
&\quad + pr\{6a^2(q^2 + 1) - ab(q - 1) + (14ac - 3b^2)(q + 1)\} \\
&\quad + r^2\{-8a^2(q - 1) + 4ab\} \\
&= (q + 1)\{-2a^2(q^2 - q + 1) + 2ab(q - 1) - 2bc + 2c^2 \\
&\quad + p^2[-ab(q^2 - q + 1) - 2ac(q - 1) + bc] \\
&\quad + pr[6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)] \\
&\quad + r^2[-8a^2(q - 1) + 4ab]\}.
\end{aligned}$$

Hence writing, first,  $a = c = 1$ ,  $b = \frac{1}{2}$ , we obtain, after some reductions,

$$\Omega = (q + 1)\{-2q(q - 1)(q^2 - \frac{1}{2}q + 1) + p^2(q - 1)(-\frac{1}{2}q - 2) + pr(6q^2 - \frac{1}{2}q - \frac{19}{2}) + r^2(-8q + \frac{9}{2})\};$$

secondly, writing  $a = c = 1$ ,  $b = -2$ , we obtain

$$\Omega = (q + 1)\{-2(q + 1)^2(q^2 - q + 1) + p^2 \cdot 2(q - 1)^2 + 2pr(3q^2 - 2q + 6) - 8r^2q\};$$

and, thirdly, writing  $a = 1$ ,  $b = -\frac{1}{2}$ ,  $c = -\frac{1}{2}$ , we obtain

$$\Omega = (q + 1)\{(-2q^3 + q^2 - \frac{1}{2}q + \frac{1}{2}) + p^2(-\frac{1}{2}q^2 + \frac{1}{2}q - \frac{1}{2}) + pr(6q^2 - \frac{19}{2}q - \frac{27}{4}) + r^2(-8q + \frac{1}{2})\}.$$

## 652.

## ON A SEXTIC TORSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 229–235.]

(Page 68.) The torse having for its edge of regression or cuspidal edge the curve defined by the equations  $x = \cos \phi$ ,  $y = \sin \phi$ ,  $z = \cos 2\phi$ , is an interesting and convenient one for the construction of a model, and it is here considered partly from that point of view.

The edge is a quadriquadric curve, the intersection of the cylinder  $x^2 + y^2 = 1$  with the parabolic hyperboloid  $z = x^2 - y^2$ ; the cylinder regarded as a cone having its vertex at infinity on the line  $x = 0$ ,  $y = 0$ , viz. the vertex is on the hyperboloid, or the curve is a nodal quadriquadric (the node being thus an isolated point at infinity on the line in question), and the torse is consequently of the order  $8 - 2$ , = 6, viz. it is a sextic torse.

The edge is a bent oval situate on the cylinder  $x^2 + y^2 = 1$ , such that, regarding  $\phi$  as the azimuth (or angle measured along the circular base from its intersection with the axis of  $x$ ), the altitude  $z$  is given by the equation  $z = \cos 2\phi$ ; viz. there are in the plane  $xz$ , or, say in the planes  $xz$ ,  $x'z$ , two maxima altitudes  $z = 1$ , and in the plane  $yz$ , or, say in the planes  $yz$  and  $y'z$ , two minima altitudes  $z = -1$ . The sections by these principal planes are, as is seen at once, nodal curves on the surface; they are, in fact, the cubic curves  $z = 3 - \frac{2}{x^2}$ , viz. here as  $x$  increases from  $\pm 1$  to  $\pm\infty$   $z$  increases from the before-mentioned value 1 to 3, and  $z = -3 + \frac{2}{y^2}$ , viz. as  $y$  increases from  $\pm 1$  to  $\pm\infty$ ,  $z$  decreases from the before-mentioned value  $-1$  to  $-3$ . The two half-sheets (which meet in the cuspidal edge) intersect each other along these nodal lines, in such wise that the section of the surface by any axial plane (plane through the line  $x = 0$ ,  $y = 0$ ) is a curve having a cusp on the cuspidal edge, and such that when the axial plane coincides with either of the principal planes  $x = 0$ ,  $y = 0$ , the

(Page 69.) two half-branches of the curve coincide together with the portions which lie outside the cylinder  $x^2 + y^2 = 1$ , in fact, the portions referred to above, of the nodal curve in the plane in question; the portions which lie inside the cylinder are acnodal or isolated curves without any real sheet through them. It may be added, in the way of general description, that the section of the surface by any cylinder  $x^2 + y^2 = c^2$  ( $c > 1$ ) is a curve of the form  $z = C \cos(2\theta \pm B)$ ,  $\theta$  the angle along the base of the cylinder from the intersection with the axis of  $x$ ;  $C$ ,  $B$  are functions of  $c$ ; viz. we have for the two half sheets respectively

$$z = C \cos(2\theta + B) \quad \text{and} \quad z = C \cos(2\theta - B),$$

each curve having thus the two maxima  $+C$ , and the two minima  $-C$ ; and the two curves intersect each other at the four points in the two principal planes respectively; viz. the points for which  $\theta = 0, 90^\circ, 180^\circ, 270^\circ$ , and  $z = C \cos B, -C \cos B, C \cos B, -C \cos B$  accordingly.

Proceeding to discuss the surface analytically, we have for the equations of a generating line

$$\frac{x - \cos \phi}{-\sin \phi} = \frac{y - \sin \phi}{\cos \phi} = \frac{z - \cos 2\phi}{-2 \sin 2\phi} = \rho \text{ suppose,}$$

or say

$$\begin{aligned} x &= \cos \phi - \rho \sin \phi, \\ y &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi, \end{aligned}$$

which equations, considering therein  $\rho$ ,  $\phi$  as arbitrary parameters, determine the surface.

Writing  $x = 0$ , we find  $y = \frac{1}{\cos \phi}$ , and then  $z = -3 + 2 \sin^2 \phi$ , viz. we have

$$x = 0, \quad z = -3 + \frac{2}{y^2}, \text{ for section in plane } yz,$$

and, similarly, writing  $y = 0$ , we find  $x = \frac{1}{\cos \phi}$ , and then  $z = 3 - 2 \cos^2 \phi$ , viz.

$$y = 0, \quad z = 3 - \frac{2}{x^2}, \text{ for section by plane } xz.$$

By what precedes, these are nodal curves, crunodal for the portions

$$(y = \pm 1 \text{ to } \pm \infty, z = -1 \text{ to } -3) \text{ and } (x = \pm 1 \text{ to } \pm \infty, z = 1 \text{ to } 3)$$

respectively, acnodal for the remaining portions  $y < \pm 1, x < \pm 1$  respectively.

Writing  $x = r \cos \theta, y = r \sin \theta$ , so that the coordinates of a point on the surface are  $r, \theta, z$ , where  $r = \sqrt{x^2 + y^2}$  is the projected distance,  $\theta$  is the azimuth from the axis of  $x$ , and  $z$  is the altitude, we have

$$\begin{aligned} r \cos \theta &= \cos \phi - \rho \sin \phi, \\ r \sin \theta &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi. \end{aligned}$$

(Page 70.) We have  $r^2 = 1 + \rho^2$ ; and thence also, if  $\tan \alpha = 2\rho, = \pm 2\sqrt{(r^2 - 1)}$ , that is,

$$\cos \alpha = \frac{1}{\sqrt{(4r^2 - 3)}}, \quad \sin \alpha = \pm \frac{2\sqrt{(r^2 - 1)}}{\sqrt{(4r^2 - 3)}},$$

then

$$z = \sqrt{(4r^2 - 3)} \cos(2\phi + \alpha),$$

showing that for a given value of  $r$  (or section by the cylinder  $x^2 + y^2 = r^2$ ) the maximum and minimum values of  $z$  are  $z = \pm \sqrt{(4r^2 - 3)}$ .

But proceeding to eliminate  $\phi$ , we find

$$\begin{aligned} r^2 \cos 2\theta &= (1 - \rho^2) \cos 2\phi - 2\rho \sin 2\phi, \\ r^2 \sin 2\theta &= 2\rho \cos 2\phi + (1 - \rho^2) \sin 2\phi, \end{aligned}$$

or multiplying these by  $1 + \rho^2$  and  $2\rho$  and adding

$$r^2 \{(1 + 3\rho^2) \cos 2\theta + 2\rho^2 \sin 2\theta\} = (1 + \rho^2)^2 (\cos 2\phi - 2\rho \sin 2\phi),$$

that is,

$$r^2 \{(3r^2 - 2) \cos 2\theta \pm 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta\} = r^4 z;$$

or, finally,

$$r^2 z = (3r^2 - 2) \cos 2\theta + 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta,$$

which is the equation of the surface in terms of the coordinates  $r, \theta, z$ .

Observing that  $(3r^2 - 2)^2 + 4(r^2 - 1)^3 = r^4(4r^2 - 3)$ , we may write

$$\begin{aligned} r^2 \sqrt{(4r^2 - 3)} \cos \beta &= 3r^2 - 2, \\ r^2 \sqrt{(4r^2 - 3)} \sin \beta &= 2(r^2 - 1)^{\frac{3}{2}}, \end{aligned}$$

and therefore also

$$\tan \beta = \frac{2(r^2 - 1)^{\frac{3}{2}}}{3r^2 - 2},$$

and the equation thus becomes

$$z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta),$$

where  $z$  is the altitude belonging to the azimuth  $\theta$  in the cylindrical section, radius  $r$ . The maxima and minima altitudes are  $\pm\sqrt{(4r^2 - 3)}$ , and these correspond to the values  $\theta = \pm\frac{1}{2}\beta$ ,  $\frac{1}{2}\pi + \frac{1}{2}\beta$ ,  $\pi + \frac{1}{2}\beta$ ,  $\frac{3}{2}\pi + \frac{1}{2}\beta$ ; it is to be further noticed that when  $r = 1$ , we have  $\beta = 0$ , but as  $r$  increases and becomes ultimately infinite,  $\beta$  increases to  $\frac{1}{2}\pi$ , that is,  $\frac{1}{2}\beta$  increases from 0 to  $\frac{1}{4}\pi$ .

It may be noticed that the surface is a peculiar kind of deformation, obtained by giving proper rotations to the several cylindrical sections of the surface  $z = \sqrt{(4r^2 - 3)} \cos 2\theta$ ; viz. in rectangular coordinates this is  $r^2 z = \sqrt{(4r^2 - 3)}(x^2 - y^2)$ , that is,

$$(x^2 + y^2)^4 z^2 - \{4(x^2 + y^2) - 3\}(x^2 - y^2)^2 = 0.$$

To obtain the equation in rectangular coordinates, we have

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 16(r^2 - 1)^3 \sin^2 2\theta = 0,$$

viz. this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 \left(1 - \frac{4x^2 y^2}{r^4}\right) - 16(r^2 - 1)^3 \frac{4x^2 y^2}{r^4} = 0,$$

(Page 71.) or, what is the same thing, it is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - \frac{4x^2 y^2}{r^4} \{4(r^2 - 1)^3 + (3r^2 - 2)^2\} = 0,$$

viz. the term in  $\{ \}$  being  $r^4(4r^2 - 3)$ , this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 4x^2 y^2(4r^2 - 3) = 0,$$

or say

$$z^2(x^2 + y^2)^2 - 2z(3x^2 + 3y^2 - 2)(x^2 - y^2) + (3x^2 + 3y^2 - 2)^2 - 4x^2 y^2(4x^2 + 4y^2 - 3) = 0.$$

This may also be written

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2 y^2(z^2 - 4x^2 - 4y^2 + 3) = 0,$$

a form which puts in evidence the nodal curves

$$x = 0, \quad zy^2 = -3y^2 + 2, \quad \text{and} \quad y = 0, \quad zx^2 = 3x^2 - 2.$$

It shows also that the quadric cone  $z^2 - 4x^2 - 4y^2 + 3 = 0$  touches the surface along the curve of intersection with the surface  $z(x^2 - y^2) - 3(x^2 + y^2) + 2 = 0$ . This is, in fact, the curve of maxima and minima of the cylindrical sections, viz. reverting to the form  $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$ , or, if for greater clearness, attending only to one sheet of the surface, we write it  $z = \sqrt{(4r^2 - 3)} \cos(2\theta - \beta)$ , we have a maximum,  $z = \sqrt{(4r^2 - 3)}$ , for  $2\theta = \beta$  (or  $2\pi + \beta$ ), giving

$$\cos 2\theta = \cos \beta = \frac{3r^2 - 2}{r^2 \sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

and a minimum,  $z = -\sqrt{(4r^2 - 3)}$ , for  $2\theta = \pi + \beta$  (or  $3\pi + \beta$ ), giving

$$\cos 2\theta = -\cos \beta = -\frac{3r^2 - 2}{r^2\sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

viz. the locus is  $z^2 = 4(r^2 - 3)$ ,  $z(x^2 - y^2) = 3r^2 - 2$ ; and for  $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$  we find the same locus, viz. the equations of the locus are

$$z^2 - 4x^2 - 4y^2 + 3 = 0, \quad z(x^2 - y^2) - 3x^2 - 3y^2 + 2 = 0,$$

as above.

To put in evidence the cuspidal edge, write for a moment  $\zeta = z - x^2 + y^2$ , the equation becomes  $\{\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 2) - 4x^2y^2\}^2 + 4x^2y^2\{\zeta^2 + 2\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 3) - 4x^2y^2\} = 0$ ; viz. this is

$$\zeta^2 r^4 + 2\zeta(x^2 - y^2)(r^2 - 1)(r^2 - 2) + (r^2 - 1)^2(r^2 - 2)^2 - 4\zeta^2 x^2 y^2 (r^2 - 1)^2 = 0,$$

or writing the last term thereof in the form

$$-[r^4 - (x^2 - y^2)^2](r^2 - 1)^2,$$

and then putting  $r^2 = 1 + U$ , the equation is

$$\zeta^2(1 + 2U + U^2) + 2\zeta U(U - 1)(x^2 - y^2) + U^2(U - 1)^2 - U^2\{(U + 1)^2 - (x^2 - y^2)^2\} = 0;$$

viz. this is

$$\{\zeta - U(x^2 - y^2)\}^2 + 2U\{\zeta^2 + \zeta U(x^2 - y^2) - 2U^2\} + 4U^2 = 0,$$

(Page 72.) showing the cuspidal edge  $\zeta = 0$ ,  $U = 0$ , viz.  $z = x^2 - y^2$ ,  $x^2 + y^2 = 1$ . Moreover, along the cuspidal edge the surface is touched by  $\zeta - U(x^2 - y^2) = 0$ , that is, by  $z - (x^2 - y^2) = 0$ ; and at the points where this tangent surface again meets the surface we have  $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$ ; viz. the surface contains upon itself the curve represented by this last equation, and  $z - (x^2 - y^2) = 0$ .

As a verification, in the form

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2y^2(z^2 - 4x^2 - 4y^2 + 3) = 0$$

of the equation of the surface, write  $z = x^2 - y^2$ . If for a moment  $x^2 + y^2 = \lambda$ ,  $x^2 - y^2 = \mu$ , then the value of  $z$  is  $z = \lambda\mu$ , and the equation becomes

$$(\lambda\mu^2 - 3\lambda + 2)^2 + (\lambda^2 - \mu^2)(\lambda^2\mu^2 - 4\lambda + 3) = 0,$$

that is,

$$\mu^2(\lambda^3 - 6\lambda^2 + 8\lambda - 3) - 4\lambda^3 + 12\lambda^2 - 12\lambda + 4 = 0;$$

or, what is the same thing,

$$(\lambda - 1)^2\{\mu^2(\lambda + 3) - 4\} = 0,$$

so that we have  $(\lambda - 1)^2 = 0$ , or else  $\mu^2(\lambda + 3) - 4 = 0$ ; viz.  $(x^2 + y^2 - 1)^2 = 0$ , or else  $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$ , agreeing with the former result.

In polar coordinates, the surface is touched along the cuspidal edge by the surface  $z = r^2 \cos 2\theta$ , and where this again meets the surface we have  $r^4(r^2 + 3) \cos^2 2\theta - 4 = 0$ .

For the model, taking the unit to be 1 inch, I suppose that for the edge of regression we have

$$x = 2 \cos \phi, \quad y = 2 \sin \phi, \quad z = 5 + (.45) \cos 2\phi$$

viz. the curve is situate on a cylinder radius 2 inches. And I construct in zinc-plate the cylindric sections, or say the templets, for one sheet of the surface, for the several radii 2, 3, ..., 8 inches; taking the radius as  $k$  inches, the circumference of the cylinder, or entire base of the flattened templet, is  $= 2k\pi$ ; and the altitude, writing  $2\theta$  in place of  $2\theta - \beta$  as above, is given by the formula  $z = 5 + (.45)\sqrt{(k^2 - 3)}\cos 2\theta$ , so that the half altitude of the wave is  $= (.45)\sqrt{(k^2 - 3)}$ ; having this value, the curve is at once constructed geometrically. We have, moreover,  $\cos \beta = \frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$ ; the numerical values then are

| $k$ | $2k\pi$ | $(.45)\sqrt{(k^2 - 3)}$ | $\frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$ | $\frac{1}{2}\beta$ |
|-----|---------|-------------------------|----------------------------------------|--------------------|
| 2   | 12.57   | 0.45                    | 1.00                                   | 0°                 |
| 3   | 18.85   | 1.10                    | .86                                    | 15                 |
| 4   | 25.13   | 1.62                    | .69                                    | 23                 |
| 5   | 31.42   | 2.11                    | .57                                    | 27 $\frac{1}{2}$   |
| 6   | 37.70   | 2.59                    | .48                                    | 30 $\frac{1}{2}$   |
| 7   | 43.98   | 3.05                    | .42                                    | 32 $\frac{1}{2}$   |
| 8   | 50.27   | 3.51                    | .36                                    | 34                 |

the altitudes in the successive templets being thus included between the limits  $5 \pm 0.45$ ,  $5 \pm 1.10$ , ...,  $5 \pm 3.51$ .



## 653.

## ON A TORSE DEPENDING ON THE ELLIPTIC FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 235–241.]

(Page 73.) On attempting to cover with paper one half-sheet of the foregoing sextic torse, [652], I found that the paper assumed approximately the form of a circular annulus of an angle exceeding  $360^\circ$ , and this led me to consider the general theory of the construction of a torse in paper, and, in particular, to consider the torsos such that when developed into a plane the edge of regression becomes a circular arc. It is scarcely necessary to remark that, to construct in paper a circular annulus of an angle exceeding  $360^\circ$ , we have only to take a complete annulus, cut it along a radius, and then insert (gumming it on to the two terminal radii) a portion of an equal circular annulus; drawing from each point of the inner circular boundary a half-tangent, and considering these half-tangents as rigid lines, the paper will bend round them so as to form the half-sheet of a torse having for its edge of regression this inner boundary, which will assume the form of a closed curve with two equal and opposite maxima and two equal and opposite minima, described on a cylinder, and being approximately such as the curve given by the equations

$$x = \cos \theta, \quad y = \sin \theta, \quad z = m \cos 2\theta.$$

Considering, in general, an arc  $PQ$  (without inflexions) of any curve, and drawing at the consecutive points  $P, P', P'', \&c.$  the several half-tangents  $PT, P'T', P''T'', \dots$ , then, considering these as rigid lines and bending the paper round them, we have the half-sheet of a torse, having for its edge of regression the curve in question now bent into a double curvature. It is, moreover, clear that the edge of regression has at each point thereof the same radius of absolute curvature as the original plane curve; in fact, if in the plane curve  $PP' = ds$ , and the angle  $T'PT$  between the consecutive half-tangents  $PT$  and  $P'T'$  be  $= d\phi$ , these quantities  $ds$  and

$$C. X. \qquad 10$$

(Page 74.)  $d\phi$  remain unaltered in the curve of double curvature; and the radius of absolute curvature is given by the equation  $\rho d\phi = ds$ . In particular when, as above, the arc is a circular one, say of radius  $= a$ , then, however the paper is bent, the edge of regression has at each point thereof the radius of absolute curvature  $= a$ .

Consider on any given surface, at a given point  $P$  thereof, and in a given direction, an element of length  $PP'$ , then (under the restrictions presently mentioned) we can determine the consecutive element  $P'P''$ , such that the curve  $PP'P'' \dots$  shall have at  $P$  a radius of absolute curvature  $= a$ ; in fact,  $r$  being the radius of curvature of the normal section of the surface through the element  $PP'$ , the radius of curvature of the section inclined at an angle  $\theta$  to the normal section is  $= r \cos \theta$ ; so that we have only to take the section at the inclination  $\theta, = \cos^{-1} \frac{a}{r}$ , to the normal section, and we have the consecutive element  $P'P''$  such that the radius of absolute curvature of the curve  $PP'P''$  is  $= a$ . The necessary restriction, of course, is that  $r > a$ ; thus, if at the given point  $P$  the two principal radii of curvature are of the same sign (to fix the ideas, let the two principal radii and also  $a$  be each of them positive), then we may on the surface determine a direction  $PQ$ , for which the radius of curvature of the normal section is  $= a$ ; and then the direction of the element  $PP'$  may be any direction between  $PQ$  and the direction  $PR$ , corresponding to the greatest of the two principal radii.

Having obtained the element  $P'P''$ , we may, if the radius of absolute curvature at  $P'$  be given, construct the next element  $P''P'''$ , and so on; that is to say, on a given surface starting from a given point  $P$  and given initial direction  $PP'$ , we can (under a restriction, as above, as to the curvature at the different points of the surface) construct a curve having at the successive points thereof given values of the radius of absolute curvature; viz., the value may be given either as

a function of the coordinates of the point on the surface, or as a function of the length of the curve measured say from the initial point  $P$ ; it is in this last manner that in what follows the value of the radius of absolute curvature is assumed to be given.

We may thus, taking on paper an arc  $PQ$  with its half-tangents, apply it to a given surface, the point  $P$  to a given point, and the infinitesimal arc  $PP'$  to an element  $PP'$  in a given direction from the given point; and we thus obtain the half-sheet of a torse having for its edge of regression a determinate curve upon the surface. In particular, the arc  $PQ$  may be circular of the radius  $a$ , and the surface be a circular cylinder of radius  $\alpha$ ; and we thus obtain the torse having for edge of regression a curve on the cylinder radius  $\alpha$ , and such that the radius of absolute curvature is at each point  $= a$ . There are three cases according as  $a > \alpha$ ,  $a = \alpha$ , or  $a < \alpha$ ; it is to be remarked that if  $a > \alpha$ , then the curve must at each point cut the generating line of the cylinder at an angle not exceeding  $\cos^{-1} \frac{\alpha}{a}$ , but that in the other two cases the angle may have any value whatever; and, further, that in every case when the angle is  $= 0$ , viz. when the curve touches a generating line of the cylinder, then the osculating plane of the curve coincides with the tangent plane of the cylinder.

(Page 75.) The analytical theory is very simple. Taking  $x, y, z$  as functions of the length  $s$ , we have

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1;$$

the condition, which expresses that the radius of absolute curvature is  $= a$ , then is

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = \frac{1}{a^2}.$$

By what precedes, the point  $(x, y, z)$  may be taken to be upon a given surface, say upon the cylinder  $x^2 + y^2 = \alpha^2$ ; we may then write  $x = \alpha \cos \theta$ ,  $y = \alpha \sin \theta$ . Taking instead of  $s$  any independent variable  $u$  whatever, and using accents to denote the derived functions in regard to  $u$ , the equations become

$$\begin{aligned} x'^2 + y'^2 + z'^2 &= s'^2, \\ x''^2 + y''^2 + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \\ x &= \alpha \cos \theta, \quad y = \alpha \sin \theta. \end{aligned}$$

From the last two equations we obtain

$$x'^2 + y'^2 = \alpha^2 \theta'^2, \quad x''^2 + y''^2 = \alpha^2 (\theta'^4 + \theta''^2),$$

and the first two equations thus become

$$\begin{aligned} \alpha^2 \theta'^2 + z'^2 &= s'^2, \\ \alpha^2 \theta'^4 + \alpha^2 \theta''^2 + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \end{aligned}$$

and from the first of these we find

$$s'' = \frac{\alpha^2 \theta' \theta'' + z' z''}{(\alpha^2 \theta'^2 + z'^2)^{\frac{1}{2}}},$$

whence the second equation is

$$\alpha^2 (\theta'^4 + \theta''^2) + z''^2 - \frac{(\alpha^2 \theta' \theta'' + z' z'')^2}{\alpha^2 \theta'^2 + z'^2} = \frac{1}{a^2} (\alpha^2 \theta'^2 + z'^2)^2,$$

or reducing, this is

$$(\alpha^2 \theta'^2 + z'^2)(\theta''^2 + \theta'^4) + (z''^2 \theta'^2 - 2\theta' \theta'' z' z'' - \alpha^2 \theta'^2 \theta''^2) = \frac{1}{a^2}(\alpha^2 \theta'^2 + z'^2)^3.$$

Taking  $\theta$  as the independent variable, we have  $\theta' = 1$ ,  $\theta'' = 0$ , and the equation becomes

$$(\alpha^2 + z'^2) + z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3,$$

or, what is the same thing,

$$z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3 - (\alpha^2 + z'^2)^2.$$

Write here

$$\alpha^2 + z'^2 = \Omega^2,$$

10—2

(Page 76.) then

$$z'' = \frac{\Omega \Omega'}{\sqrt{(\Omega^2 - \alpha^2)}},$$

and the equation becomes

$$\frac{\Omega'^2}{\Omega^2 - \alpha^2} = \frac{\Omega^4}{a^2 \alpha^2} - 1,$$

or say

$$\frac{a\alpha d\Omega}{\sqrt{(\Omega^2 - \alpha^2) \cdot \Omega^4 - a^2 \alpha^2}} = d\theta,$$

viz. this equation determines  $\Omega$  as a function of  $\theta$ , and we then have

$$\begin{cases} ds = \Omega d\theta, \\ dz = \sqrt{(\Omega^2 - \alpha^2)} d\theta, \\ x = \alpha \cos \theta, \\ y = \alpha \sin \theta, \end{cases}$$

equations which determine  $x$ ,  $y$ ,  $z$ ,  $s$  as functions of the parameter  $\theta$ , and give thus the edge of regression of the torse in question.

It is clear that the formulae are very much simplified in the case  $a = \alpha$ , where the radius of absolute curvature  $a$  is equal to the radius  $\alpha$  of the cylinder; but it is worth while to develop the general case somewhat further.

Considering the elliptic functions  $\text{sn } u$ ,  $\text{cn } u$ ,  $\text{dn } u$ , to the modulus  $k (= k') = \frac{1}{\sqrt{2}}$ , assume

$$\Omega = \frac{\sqrt{(a\alpha)} \text{ dn } u}{k \text{ sn } u},$$

then

$$\begin{aligned} d\Omega &= -\frac{\sqrt{(a\alpha)} \text{ cn } u \text{ dn } u}{k \text{ sn}^2 u}, \\ \Omega^2 - \alpha^2 &= \frac{a\alpha}{k^2 \text{ sn}^2 u} \left( \text{dn}^2 u - \frac{\alpha}{a} k^2 \text{ sn}^2 u \right), \\ &= \frac{a\alpha}{k^2 \text{ sn}^2 u} \left\{ 1 - \left( 1 + \frac{\alpha}{a} \right) k^2 \text{ sn}^2 u \right\}, \\ \Omega^4 - a^2 \alpha^2 &= \frac{a^2 \alpha^2}{k^4 \text{ sn}^4 u} (\text{dn}^4 u - k^4 \text{ sn}^4 u), \end{aligned}$$

$$= \frac{a^2 \alpha^2}{k^4 \operatorname{sn}^4 u} (1 - 2k^2 \operatorname{sn}^2 u), \quad = \frac{a^2 \alpha^2}{k^4 \operatorname{sn}^4 u} \operatorname{cn}^2 u,$$

and hence

$$d\theta = \frac{k^2 \operatorname{sn} u \, du}{\sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \operatorname{sn}^2 u\}}},$$

$$dz = \frac{k \sqrt{(a\alpha)} \, \operatorname{dn} u \, du}{\sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \operatorname{sn}^2 u\}}},$$

(Page 77.) We have thus  $z = k\sqrt{(a\alpha)} u$ , no constant of integration being required, viz.  $u$  is a mere constant multiple of  $z$ : and the first and second equations then give  $s$  and  $\theta$  as functions of  $u$ , that is, of  $z$ ; but it is obviously convenient to retain  $u$  instead of expressing it in terms of  $z$ . As regards the form of these integrals observe that, writing  $\operatorname{sn} u = \lambda$ , we have

$$du = \frac{d\lambda}{\operatorname{dn} u \cdot \operatorname{cn} u},$$

and thence

$$d\theta = \frac{k^2 \lambda \, d\lambda}{\operatorname{dn} u \cdot \operatorname{cn} u \cdot \sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \lambda^2\}}},$$

$$ds = \frac{k \sqrt{(a\alpha)} \, d\lambda}{\operatorname{cn} u \cdot \sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \lambda^2\}}},$$

each of which is in fact reducible to elliptic integrals, but I do not further pursue this general case.

In the particular case  $a = \alpha$ , we have

$$1 - \left(1 + \frac{\alpha}{a}\right) k^2 \operatorname{sn}^2 u = \operatorname{cn}^2 u,$$

and the equations become

$$d\theta = \frac{k^2 \operatorname{sn} u \, du}{\operatorname{cn} u}, \quad ds = \frac{k\alpha \operatorname{dn} u \, du}{\operatorname{cn} u},$$

which admit of immediate integration; viz. we have

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'},$$

or determining the constant so that  $\theta$  may vanish for  $u = 0$ , say

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right),$$

and

$$s = \frac{1}{2} k\alpha \log \left( \frac{1 + \operatorname{sn} u}{1 - \operatorname{sn} u} \right);$$

viz. to verify these results we have

$$\frac{d\theta}{du} = \frac{1}{2} \frac{k^2}{k'} \cdot \frac{[\operatorname{dn} u + k']'(\operatorname{dn} u - k') - (\operatorname{dn} u + k')[\operatorname{dn} u - k']'}{\operatorname{dn}^2 u - k'^2} = -\frac{k^2 \operatorname{sn} u \operatorname{cn} u}{\operatorname{dn}^2 u - k'^2} = k^2 \frac{\operatorname{sn} u}{\operatorname{cn} u},$$

and

$$\frac{ds}{du} = \frac{1}{2} k\alpha \left( \frac{1}{1 + \operatorname{sn} u} + \frac{1}{1 - \operatorname{sn} u} \right) \cdot \operatorname{cn} u \operatorname{dn} u = \frac{k\alpha \operatorname{cn} u \operatorname{dn} u}{1 - \operatorname{sn}^2 u} = \frac{k\alpha \operatorname{dn} u}{\operatorname{cn} u}.$$

(Page 78.) Hence, recurring to the original equations, and writing for convenience  $a = \alpha = 1$ , we see that a solution of the simultaneous equations

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1,$$

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = 1,$$

is

$$x = \cos \theta, \quad y = \sin \theta, \quad z = ku,$$

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right), \quad s = \frac{1}{2} k \log \left( \frac{1 + \operatorname{sn} u}{1 - \operatorname{sn} u} \right),$$

where, as before,  $k = k' = \frac{1}{\sqrt{2}}$ .

Restoring the radius  $\alpha$ , and writing the system in the form

$$x = \alpha \cos \theta, \quad y = \alpha \sin \theta, \quad z = k\alpha u,$$

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right), \quad s = \frac{1}{2} k\alpha \log \left( \frac{1 + \operatorname{sn} u}{1 - \operatorname{sn} u} \right),$$

we see that, as  $u$  passes from  $u = 0$  to  $u = K$ , and therefore  $z$  from  $z = 0$  to  $z = k\alpha K$  ( $K$  the complete function  $\int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}$ ), then  $\theta$  and  $s$  each pass from 0 to  $\infty$ ; and, similarly, as  $u$  passes from  $u = 0$  to  $u = -K$ , that is, as  $z$  passes from 0 to  $-k\alpha K$ , then  $\theta$  passes from 0 to  $\infty$ , and  $s$  from  $s = 0$  to  $s = -\infty$ ; viz. the curve makes in each direction an infinity of revolutions about the cylinder. Developing the cylinder,  $\alpha\theta$  becomes an  $x$ -coordinate; viz. we have thus the plane curve

$$z = k\alpha u, \quad x = \frac{1}{2} \frac{k^2 \alpha}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right),$$

which is a curve extending from the origin in the direction  $x$  positive, to touch at infinity the two parallel asymptotes  $z = \pm k\alpha K$ ; and conversely, when such a plane curve is wound about the cylinder, there will be in each direction an infinity of revolutions round the cylinder.

## 654.

## ON CERTAIN OCTIC SURFACES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 249–264.]

(Page 79.) I consider the torse generated by the tangents of the quadriquadric curve, the intersection of the two quadric surfaces

$$\begin{aligned} ax^2 + by^2 + cz^2 + dw^2 &= 0, \\ a'x^2 + b'y^2 + c'z^2 + d'w^2 &= 0; \end{aligned}$$

then, writing

$$\begin{aligned} bc' - b'c &= a', & ad' - a'd &= f', \\ ca' - c'a &= b', & bd' - b'd &= g', \\ ab' - a'b &= c', & cd' - c'd &= h', \end{aligned}$$

and therefore

$$a'f' + b'g' + c'h' = 0,$$

the equation of the torse, writing for greater convenience  $(a, b, c, f, g, h)$  in place of  $(a', b', c', f', g', h')$ , but understanding these letters as signifying the accented letters  $(a', b', c', f', g', h')$ , is

$$\begin{aligned} &a^2 f^2 y^2 z^4 + b^2 g^2 z^2 x^4 + c^2 h^2 x^2 y^4 \\ &+ a^2 f^2 z^2 w^4 + b^2 g^2 x^2 w^4 + c^2 h^2 y^2 w^4 \\ &+ 2b^2 d' g h x^2 y^2 z^2 - 2c^2 f^2 a h x^4 y^2 w^2 + b^2 f^2 a g x^4 z^2 w^2 \\ &+ 2a^2 h^2 f g z^2 w^2 x^2 - 2a^2 h f b f y^2 z^2 w^2 + 2c^2 g^2 b h f x^2 w^2 \\ &+ 2a^2 g^2 f g z^2 w^2 x^2 - 2b^2 h^2 c g z^2 x^2 w^2 + 2a^2 h^2 c f z^2 y^2 w^2 \\ &- 2b c g^2 h^2 z^2 w^2 x^2 - 2c a h^2 f^2 z^2 w^2 x^2 - 2a b f^2 g^2 w^4 x^2 y^2 \\ &+ 2(bg - ch)(ch - af)(af - bg)x^2 y^2 z^2 w^2 = 0. \end{aligned}$$

(Page 80.) If in this equation we write

$$\begin{aligned} a^2 f^2 &= a, & b^2 g^2 &= b, & c^2 h^2 &= c, \\ a'^2 f'^2 &= f, & b'^2 g'^2 &= g, & c'^2 h'^2 &= h; \end{aligned}$$

and therefore

$$\begin{aligned} bc' - b'c &= \frac{a}{\sqrt{(af)}}, & ad' - a'd &= \frac{f}{\sqrt{(af)}}, \\ ca' - c'a &= \frac{b}{\sqrt{(bg)}}, & bd' - b'd &= \frac{g}{\sqrt{(bg)}}, \\ ab' - a'b &= \frac{c}{\sqrt{(ch)}}, & cd' - c'd &= \frac{h}{\sqrt{(ch)}}, \end{aligned}$$

and consequently

$$(af)^{\frac{1}{2}} + (bg)^{\frac{1}{2}} + (ch)^{\frac{1}{2}} = 0;$$

then the equation becomes

$$\begin{aligned}
 & a^2y^2z^4 + b^2z^2x^4 + c^2x^2y^4 \\
 & + f^2x^2w^4 + g^2y^2w^4 + h^2z^2w^4 \\
 & + 2bcx^4y^2z^2 - 2cfx^4y^2w^2 + 2bfx^4z^2w^2 \\
 & + 2cay^4z^2x^2 - 2agy^4z^2w^2 + 2cgy^4x^2w^2 \\
 & + 2ahz^4x^2y^2 - 2bhz^4x^2w^2 + 2ahz^4y^2w^2 \\
 & - 2ghw^4y^2z^2 - 2hfw^4z^2x^2 - 2fgw^4x^2y^2 \\
 & + 2\{(bg)^{\frac{1}{2}} - (ch)^{\frac{1}{2}}\}\{(ch)^{\frac{1}{2}} - (af)^{\frac{1}{2}}\}\{(af)^{\frac{1}{2}} - (ch)^{\frac{1}{2}}\}x^2y^2z^2w^2 = 0.
 \end{aligned}$$

This same equation, without the relation

$$(af)^{\frac{1}{2}} + (bg)^{\frac{1}{2}} + (ch)^{\frac{1}{2}} = 0,$$

and with an arbitrary coefficient for  $x^2y^2z^2w^2$ ; or say, the equation

$$\begin{aligned}
 & a^2y^2z^4 + b^2z^2x^4 + c^2x^2y^4 \\
 & + f^2x^2w^4 + g^2y^2w^4 + h^2z^2w^4 \\
 & + 2bcx^4y^2z^2 - 2cfx^4y^2w^2 + 2bfx^4z^2w^2 \\
 & + 2cay^4z^2x^2 - 2agy^4z^2w^2 + 2cgy^4x^2w^2 \\
 & + 2ahz^4x^2y^2 - 2bhz^4x^2w^2 + 2ahz^4y^2w^2 \\
 & - 2ghw^4y^2z^2 - 2hfw^4z^2x^2 - 2fgw^4x^2y^2 \\
 & + 2kx^2y^2z^2w^2 = 0,
 \end{aligned}$$

where  $a, b, c, f, g, h, k$  are arbitrary coefficients, is the general equation of an octic surface having the four nodal curves

$$\begin{aligned}
 x = 0, & \quad hz^2w^2 - gw^2y^2 + ay^2z^2 = 0, \\
 y = 0, & \quad -hz^2w^2 + fw^2x^2 + bz^2x^2 = 0, \\
 z = 0, & \quad gw^2x^2 - fw^2y^2 + cx^2y^2 = 0, \\
 w = 0, & \quad -ay^2z^2 - bz^2x^2 - cx^2y^2 = 0.
 \end{aligned}$$